

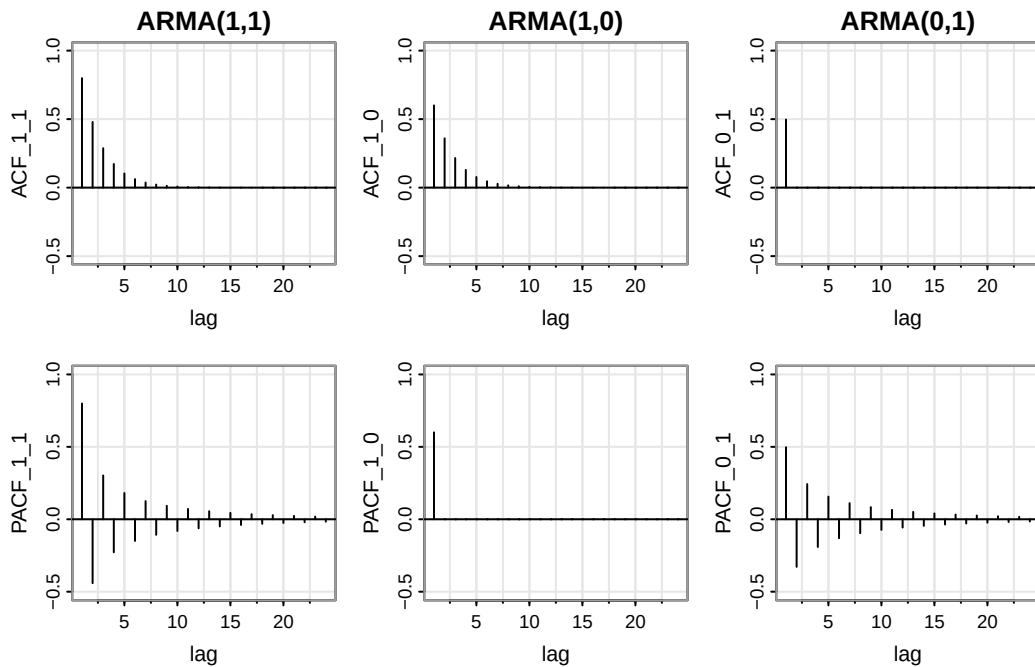
Homework 5

Jack Cunningham

I.

```
library(astsa)
```

4.4)

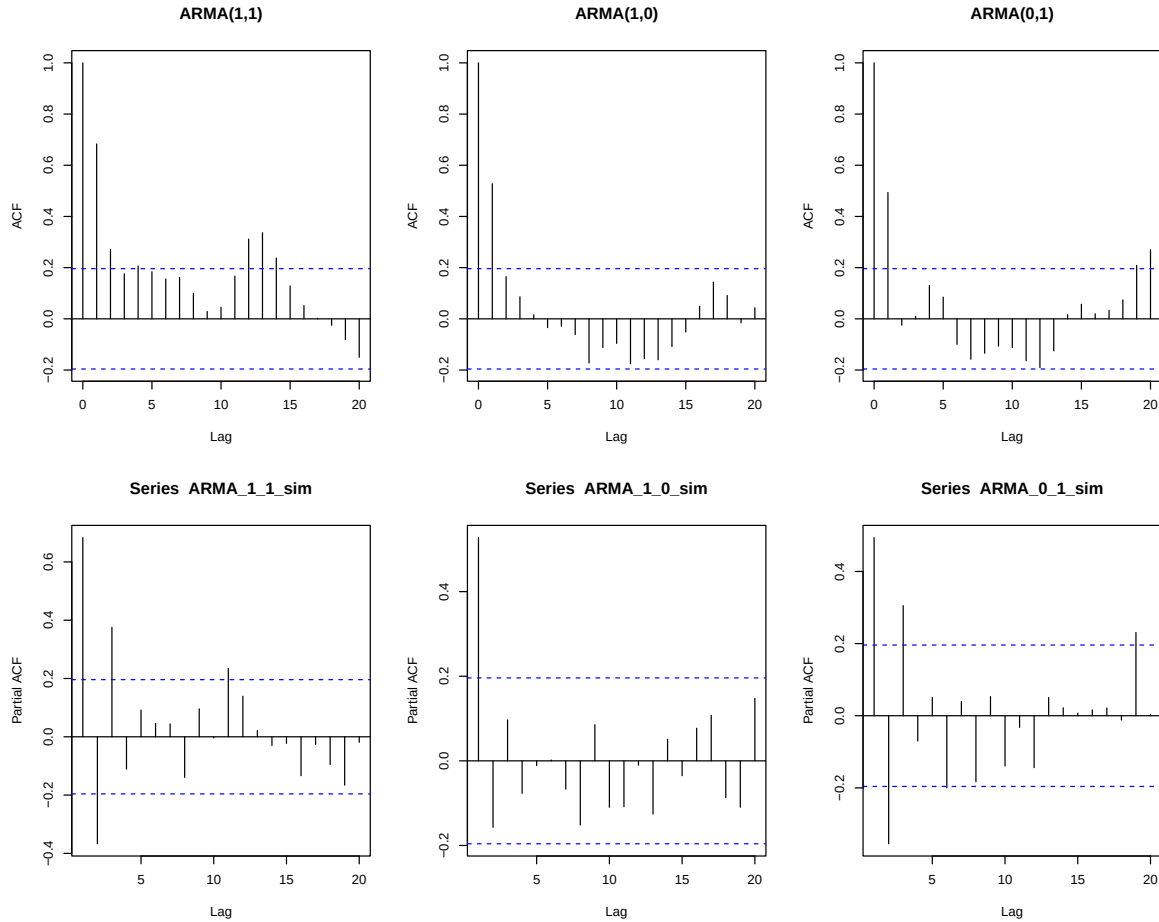


The ARMA(1,1) model has an ACF plot that tends toward 0 over time and a PACF that alternates positive and negative that tends toward zero.

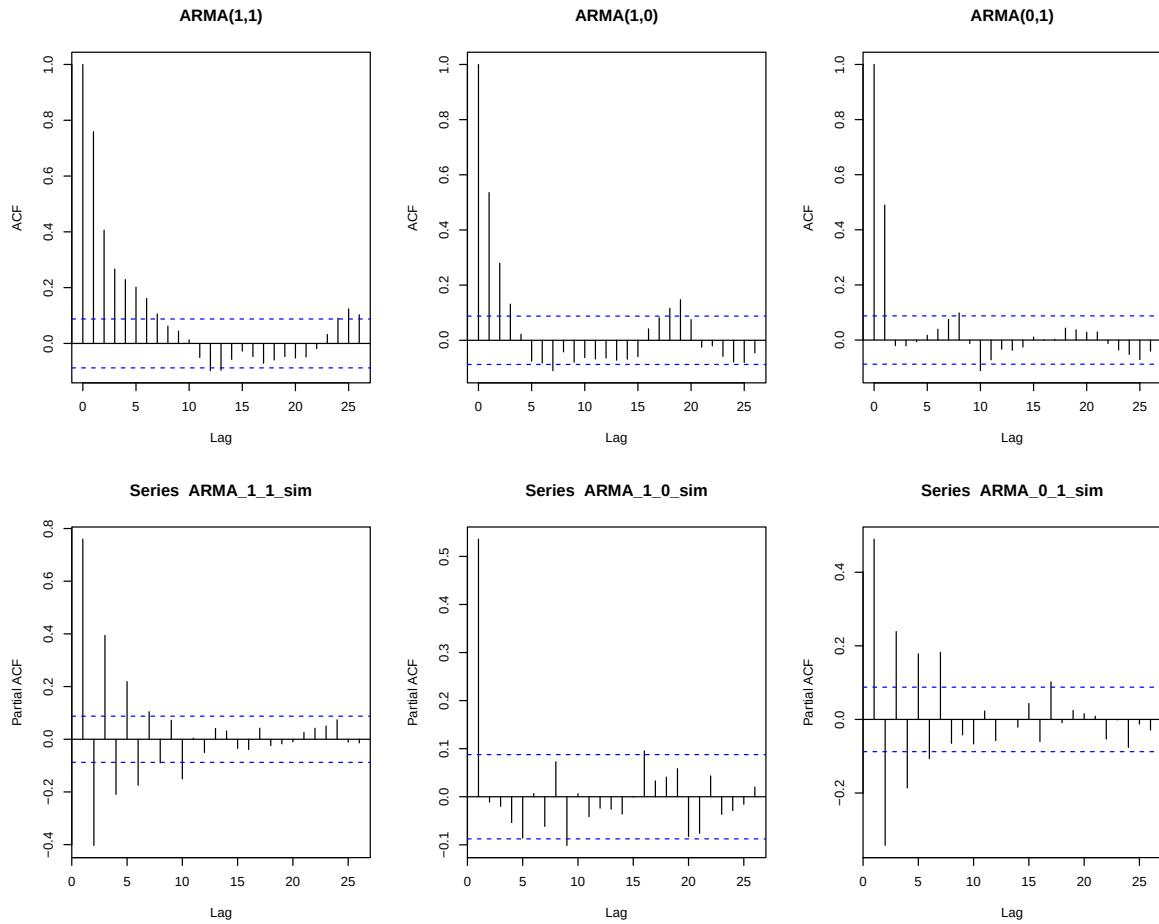
The ARMA(1,0) model has a similar ACF plot as ARMA(1,1), tending to 0 over time but with a smaller magnitude at all lags. The PACF that has correlation at lag 1 but none at later lags.

The ARMA(0,1) model has a ACF plot that has a positive value at lag 1 but none at further lags. the PACF has alternating positive and negative correlations that tend toward zero but at a lower magnitude when compared to ARMA(1,1).

Recognizing these patterns would allow us to choose the correct model from sample data.



The sample ACF and PACFs are pretty similar to the theoretical values, there are a few cases where there are lags that are significant that we wouldn't expect. In particular ARMA(1,0) has lags that are significant in the negative direction in lag 15 that we wouldn't expect as it should tend to zero. The same idea can be seen in AR(0,1) we expect all those lags to be zero past lag 1 but we see significance in lag 17.

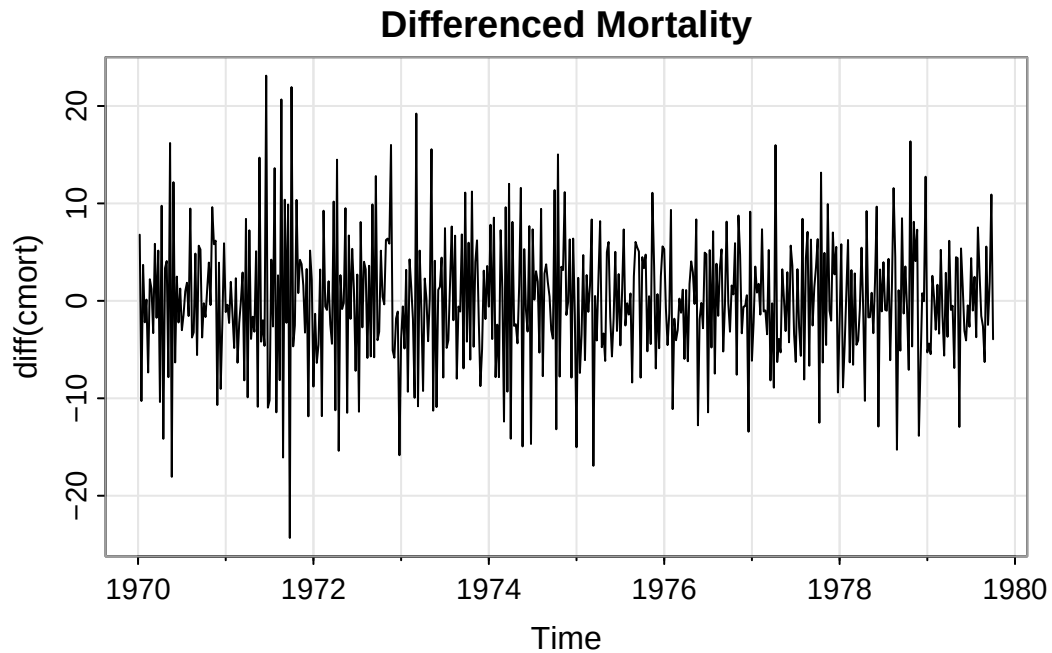


With $n = 500$ the patterns are closer to the theoretical values but we see more correlations poking into significance, likely due to the increased power from increasing n .

4.5)

a)

```
tsplot(diff(cmort), main = "Differenced Mortality")
```



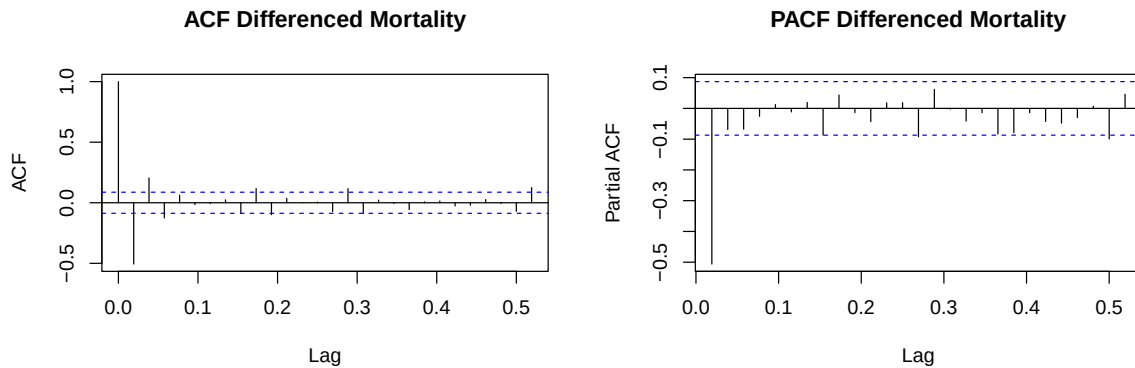
The original mortality data appears to have a small downward trend, looking like a random walk with drift.

The differenced mortality data looks like white noise.

Differencing seems reasonable because when we are dealing with a random walk with drift situation the operation yields a stationary process.

b)

```
par(mfrow = c(1,2))  
acf(diff(cmort), main = "ACF Differenced Mortality")  
pacf(diff(cmort), main = "PACF Differenced Mortality")
```



The ACF starts large and then tails to zero over time. The PACF at lag one is about -0.5 and then cuts off.

Using table 4.1, we can see that AR(1) makes sense for this data as the ACF tails off and the PACF cuts off after the first lag.

c)

Call:

```
arima(x = xdata, order = c(p, d, q), seasonal = list(order = c(P, D, Q), period = S),
      xreg = xmean, include.mean = FALSE, transform.pars = trans, fixed = fixed,
      optim.control = list(trace = trc, REPORT = 1, reltol = tol))
```

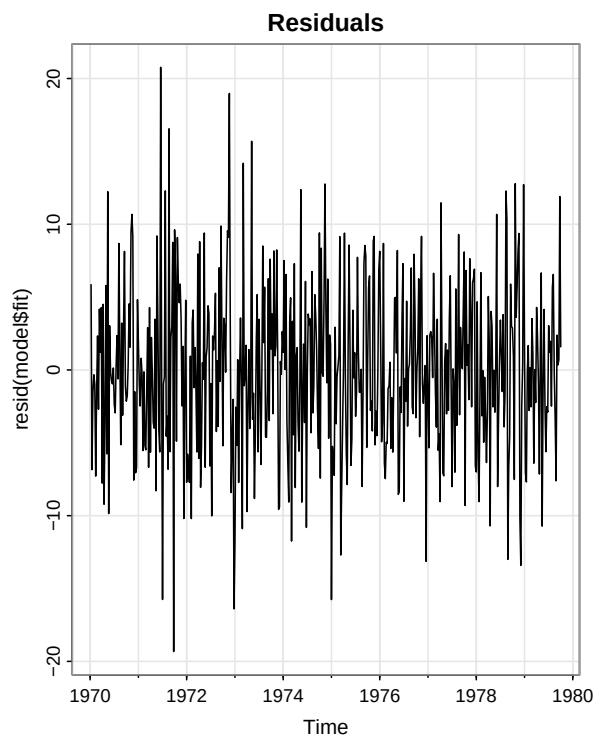
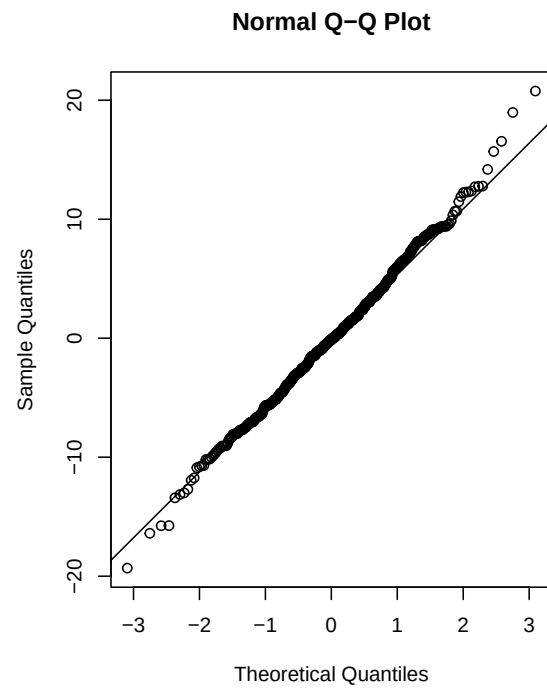
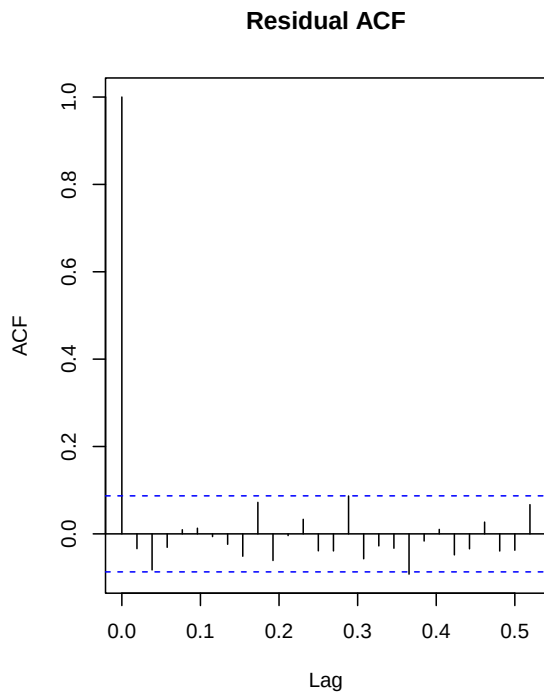
Coefficients:

```
      ar1
      -0.5064
s.e.    0.0383
```

sigma² estimated as 33.81: log likelihood = -1612.07, aic = 3228.13

The AR1 coefficient is clearly significant. The white noise variance is estimated at 33.81.

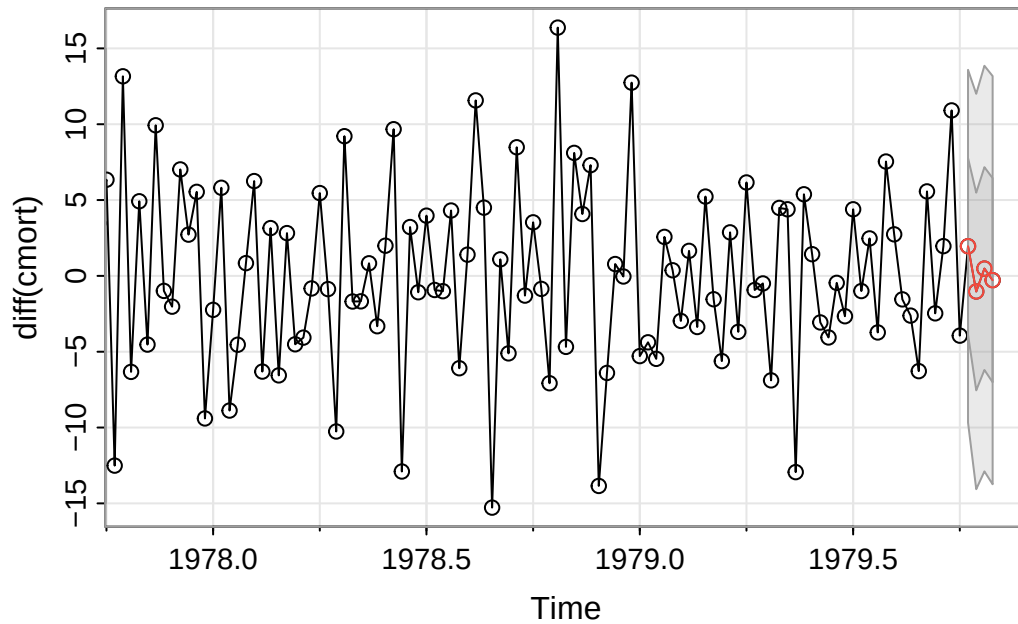
d)



We can see the residual ACF plot doesn't show any significant lags, the residuals appear

normally distributed and the residual time series plot looks like white noise.

e)



```
$pred
Time Series:
Start = c(1979, 41)
End = c(1979, 44)
Frequency = 52
[1] 1.9555424 -1.0298842 0.4818973 -0.2836494
```

```
$se
Time Series:
Start = c(1979, 41)
End = c(1979, 44)
Frequency = 52
[1] 5.814551 6.517559 6.685932 6.728429
```

f)

The forecast and prediction interval is calculated through the m-step ahead forecast formula seen in formula 4.37. That is:

$$x_{n+m}^n = \phi^m x_n$$

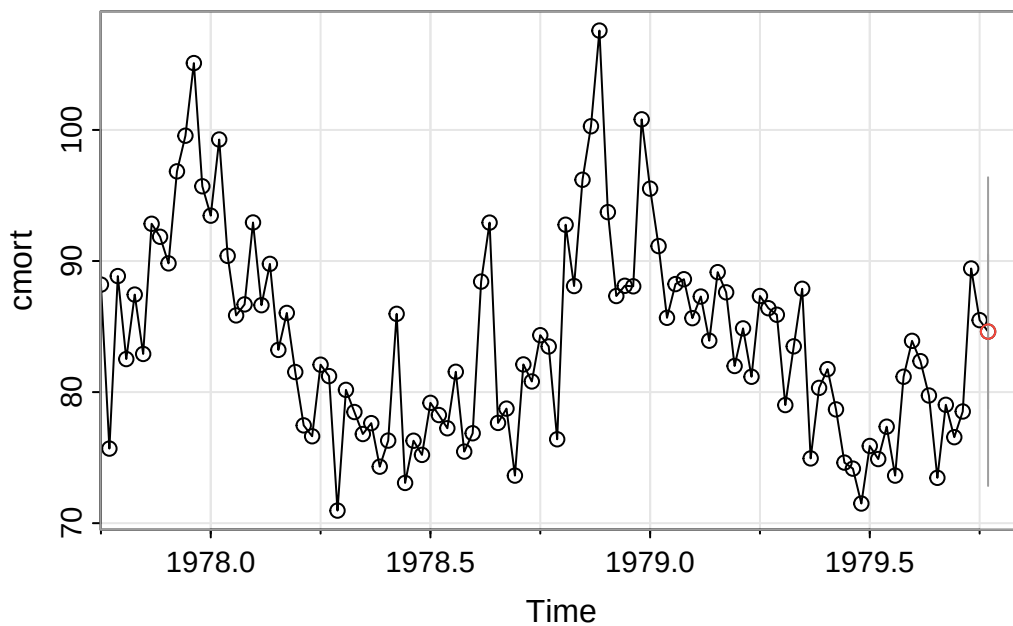
This has the prediction error seen in formula 4.38:

$$P_{n+m}^n = \sigma_w^2(1 + \phi^2 + \dots + \phi^{2(m-1)})$$

g)

Using the SARIMA forecast function with $(p = 0, d = 1, q = 1)$ we can see the forecast is 85.46562.

```
sarima.for(cmort, n.ahead = 1, p = 0, d = 1, q = 1)
```



```
$pred
Time Series:
Start = c(1979, 41)
End = c(1979, 41)
Frequency = 52
[1] 84.61435
```

```
$se
```


Time Series:
Start = c(1979, 41)
End = c(1979, 41)
Frequency = 52
[1] 5.887102

4.6)

With an AR(1) model we have:

$$x_n = \phi x_{n-1} + w_n$$

The step ahead forecast is the expected value of $E[x_{n+m}]$. We start with $m = 1$:

$$E[x_{n+1}] = E[\phi x_n + w_{n+1}] = \phi x_n = x_{n+1}^n$$

Then with $m = 2$:

$$E[x_{n+2}] = E[\phi x_{n+1} + w_{n+2}] = \phi E[x_{n+1}]$$

And we know that $E[x_{n+1}] = \phi x_n = x_{n+1}^n$ so:

$$E[x_{n+2}] = \phi^2 x_n = x_{n+2}^n$$

And we can say that generally we have:

$$x_{n+m}^n = E[x_{n+m}] = \phi^m x_n$$

Then we are tasked to show:

$$E[(x_{n+m} - x_{n+m}^n)^2] = \sigma_w^2 \frac{1 - \phi^{2m}}{1 - \phi^2}$$

We can write x_{n+m} as:

$$x_{n+m} = \phi^m x_n + \sum_{i=0}^{m-1} \phi^i w_{n+m-i}$$

Which we can substitute into $E[(x_{n+m} - x_{n+m}^n)^2]$:

$$E[(x_{n+m} - x_{n+m}^n)^2] = E[(\phi^m x_n + \sum_{i=0}^{m-1} \phi^i w_{n+m-i} - \phi^m x_n)^2] = E[(\sum_{i=0}^{m-1} \phi^i w_{n+m-i})^2]$$

Since $E[\sum_{i=0}^{m-1} \phi^i w_{n+m-i}] = 0$, we just need to find the variance of $\sum_{i=0}^{m-1} \phi^i w_{n+m-i}$:

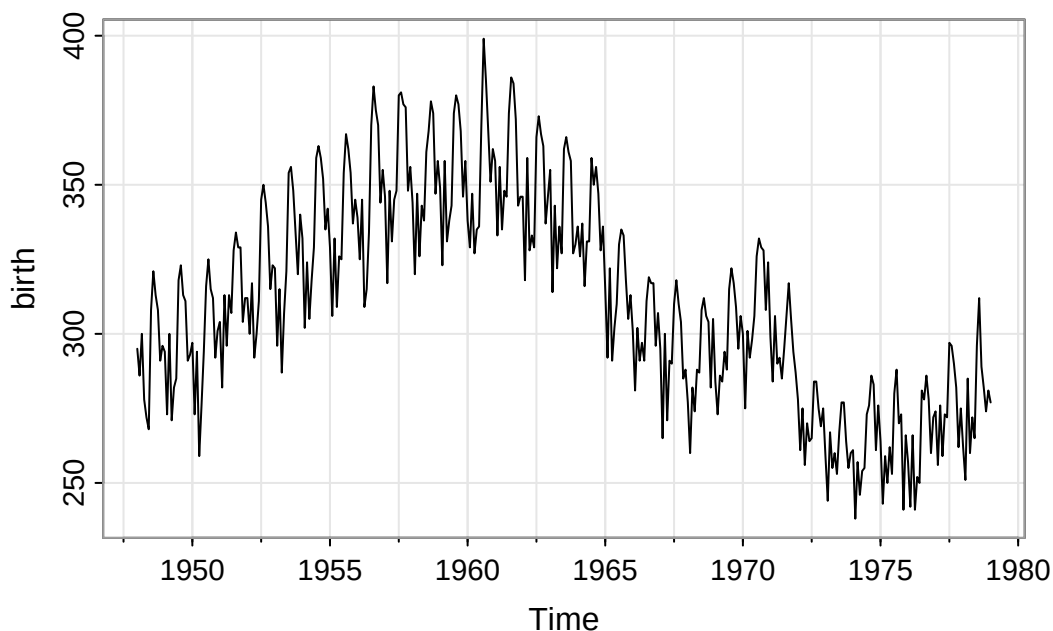
$$Var(\sum_{i=0}^{m-1} \phi^i w_{n+m-i}) = \sum_{i=0}^{m-1} \phi^{2i} \sigma_w^2$$

Which is a geometric series, so it equals

$$= \sigma_w^2 \left(\frac{1 - \phi^{2m}}{1 - \phi^2} \right)$$

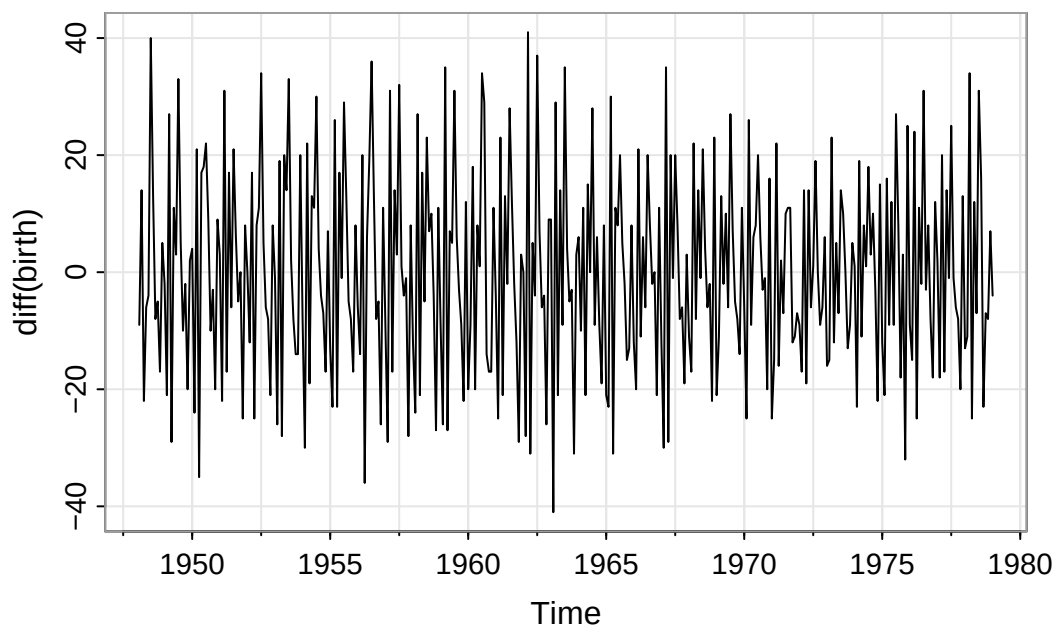
5.11)

We start our model process by investigating the time series plot:

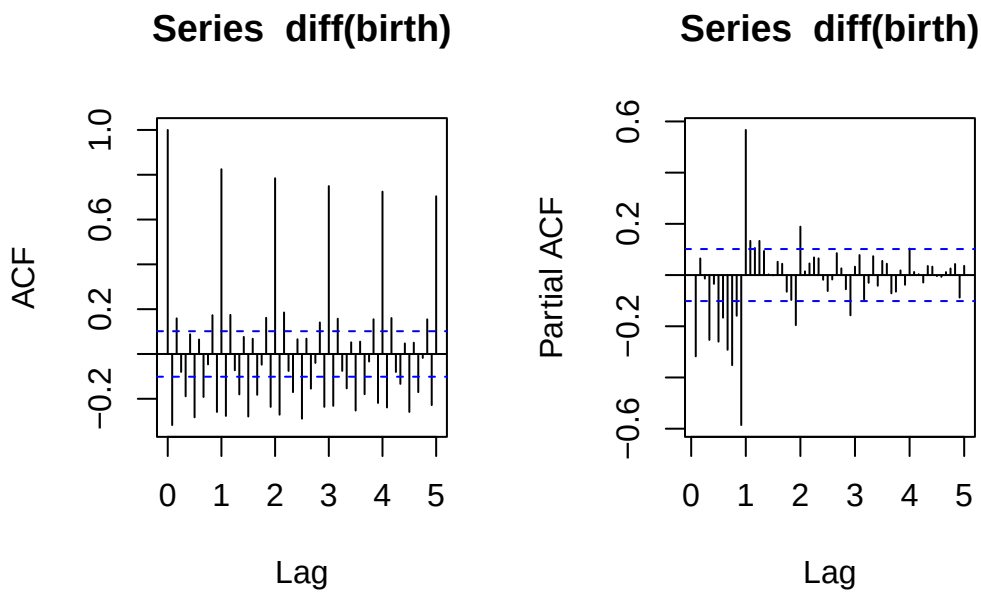


This plot looks a lot like a random walk process. Variance appears constant so no issues there. It looks like there are seasonal effects.

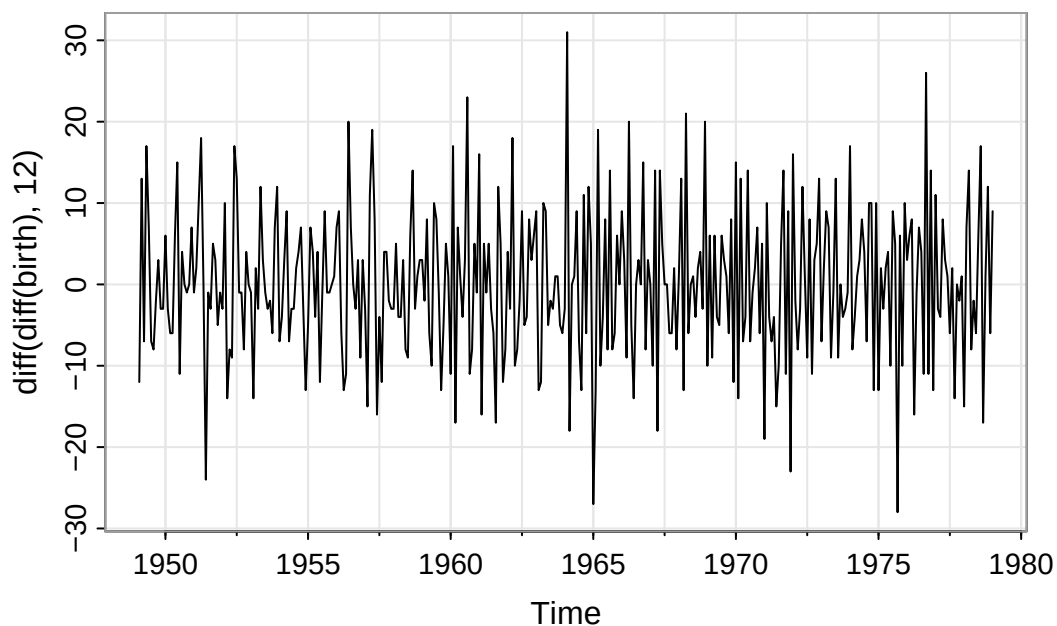
I first try a difference to deal with the random walk process.



This looks a lot closer to stationarity. To assess seasonality and choose p, q and P, Q we look at the ACF and PACF of the differenced data.

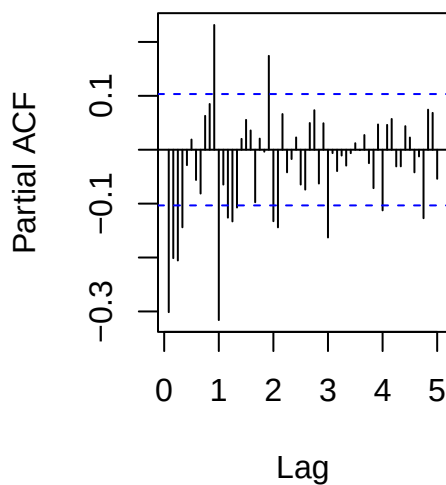
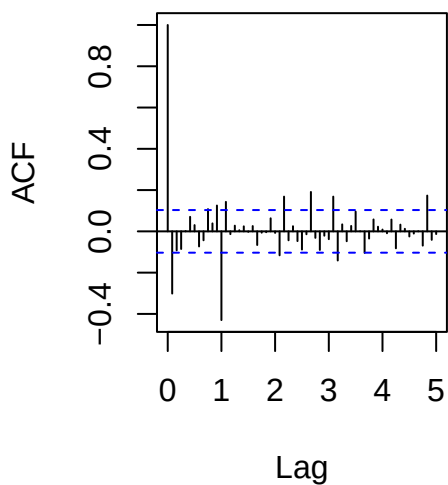


We see significant correlation between values a year apart. To go after this we different again using x_{t-12} :



Series $\text{diff}(\text{diff}(\text{birth}), 12)$

Series $\text{diff}(\text{diff}(\text{birth}), 12)$

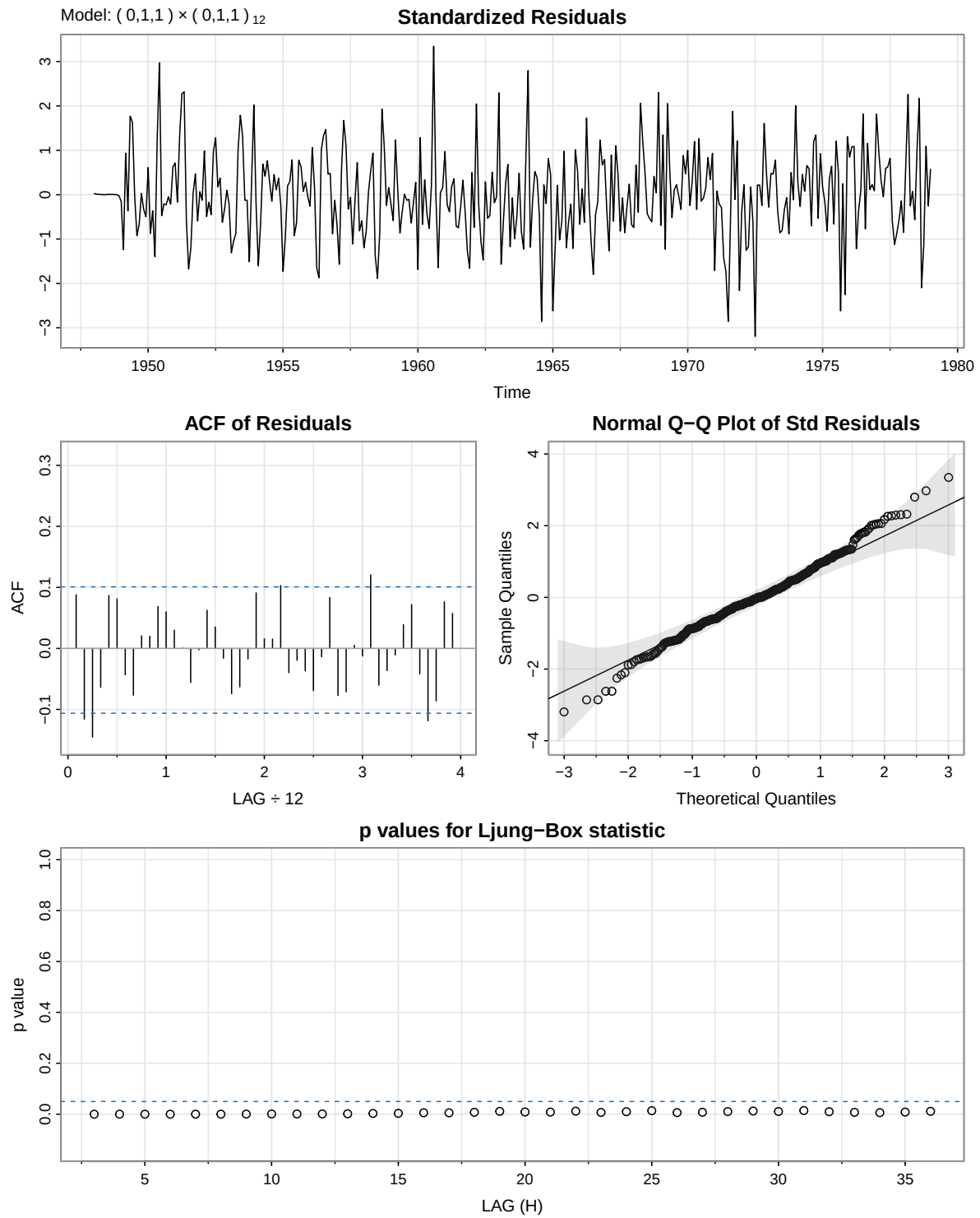


Now consulting the 5.1 table, we see that the ACF cuts off after the first 12 month lag and the PACF seems to tail off in 12 month intervals.

This would seem to indicate a $P = 0, Q = 1$ for the seasonal component of the $ARIMA(p, d, q) \times (P, D, Q)_s$.

σ^2 estimated as 47.40199 on 358 degrees of freedom

AIC = 6.74599 AICc = 6.746084 BIC = 6.778375



The $ARIMA(0,1,1) \times (0,1,1)_{12}$ model's biggest weakness is that there is still autocorrelation

remaining in the residuals. We see this most clearly in the Ljung - Box statistic where the null hypothesis of white noise is rejected at each lag.

Because of this I consider the other model $ARIMA(4, 1, 0) \times (0, 1, 1)_{12}$:

```
initial    value 2.213952
iter      2 value 1.994086
iter      3 value 1.955325
iter      4 value 1.934066
iter      5 value 1.933651
iter      6 value 1.918205
iter      7 value 1.915371
iter      8 value 1.914279
iter      9 value 1.914124
iter     10 value 1.914112
iter     11 value 1.914110
iter     11 value 1.914110
iter     11 value 1.914110
final     value 1.914110
converged

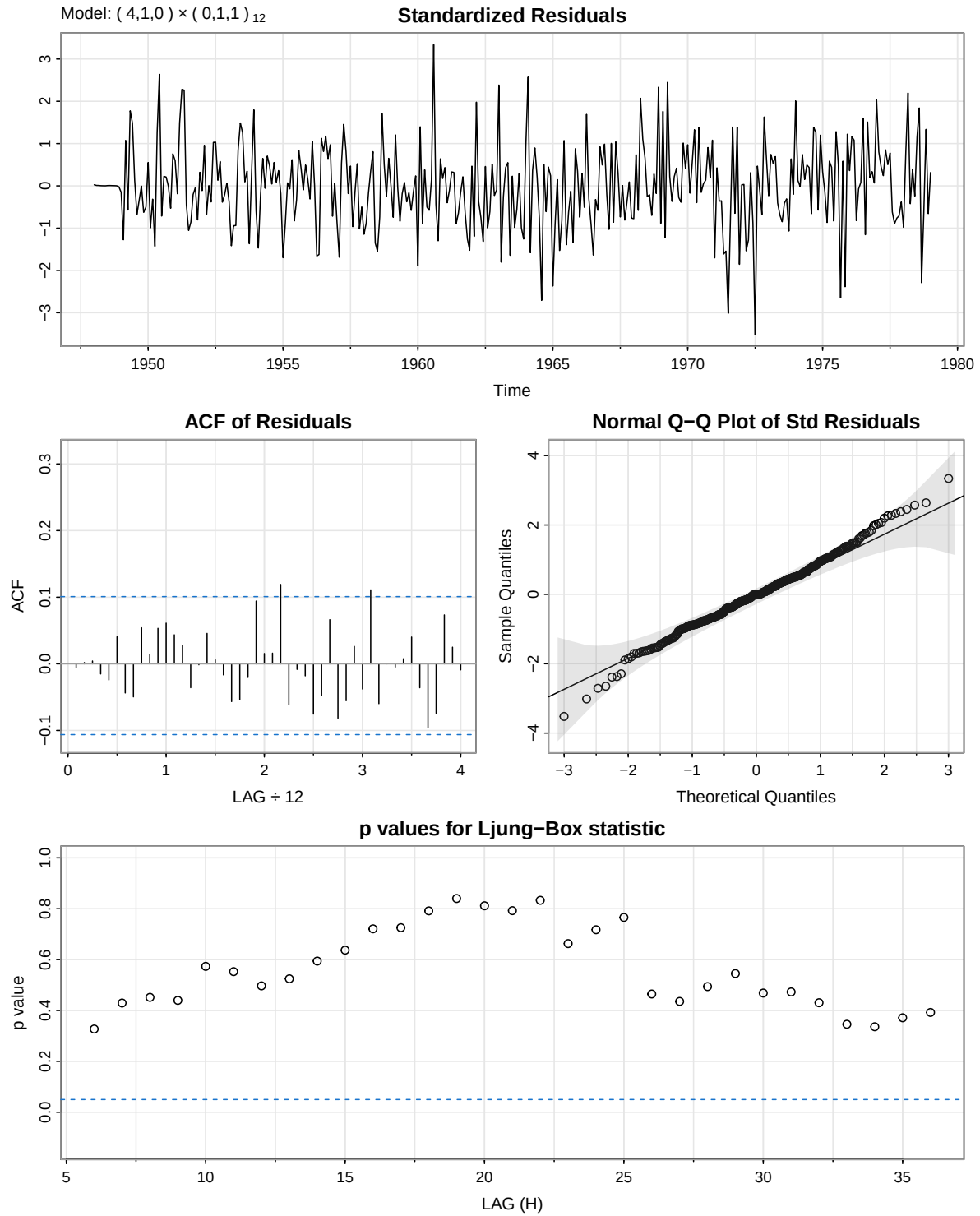
initial    value 1.923825
iter      2 value 1.922837
iter      3 value 1.922062
iter      4 value 1.921993
iter      5 value 1.921989
iter      5 value 1.921989
iter      5 value 1.921989
final     value 1.921989
converged
```

Coefficients:

	Estimate	SE	t.value	p.value
ar1	-0.3875	0.0529	-7.3207	0.000
ar2	-0.2877	0.0550	-5.2315	0.000
ar3	-0.2502	0.0544	-4.6018	0.000
ar4	-0.1747	0.0528	-3.3070	0.001
sma1	-0.8060	0.0431	-18.6959	0.000

σ^2 estimated as 45.06126 on 355 degrees of freedom

AIC = 6.715188 AICc = 6.715659 BIC = 6.779956



This model looks a lot better. Variance looks constant, the ACF plot and Ljung- Box statistic

plots support the assumption of white noise errors and residuals look normal in the qq-plot.

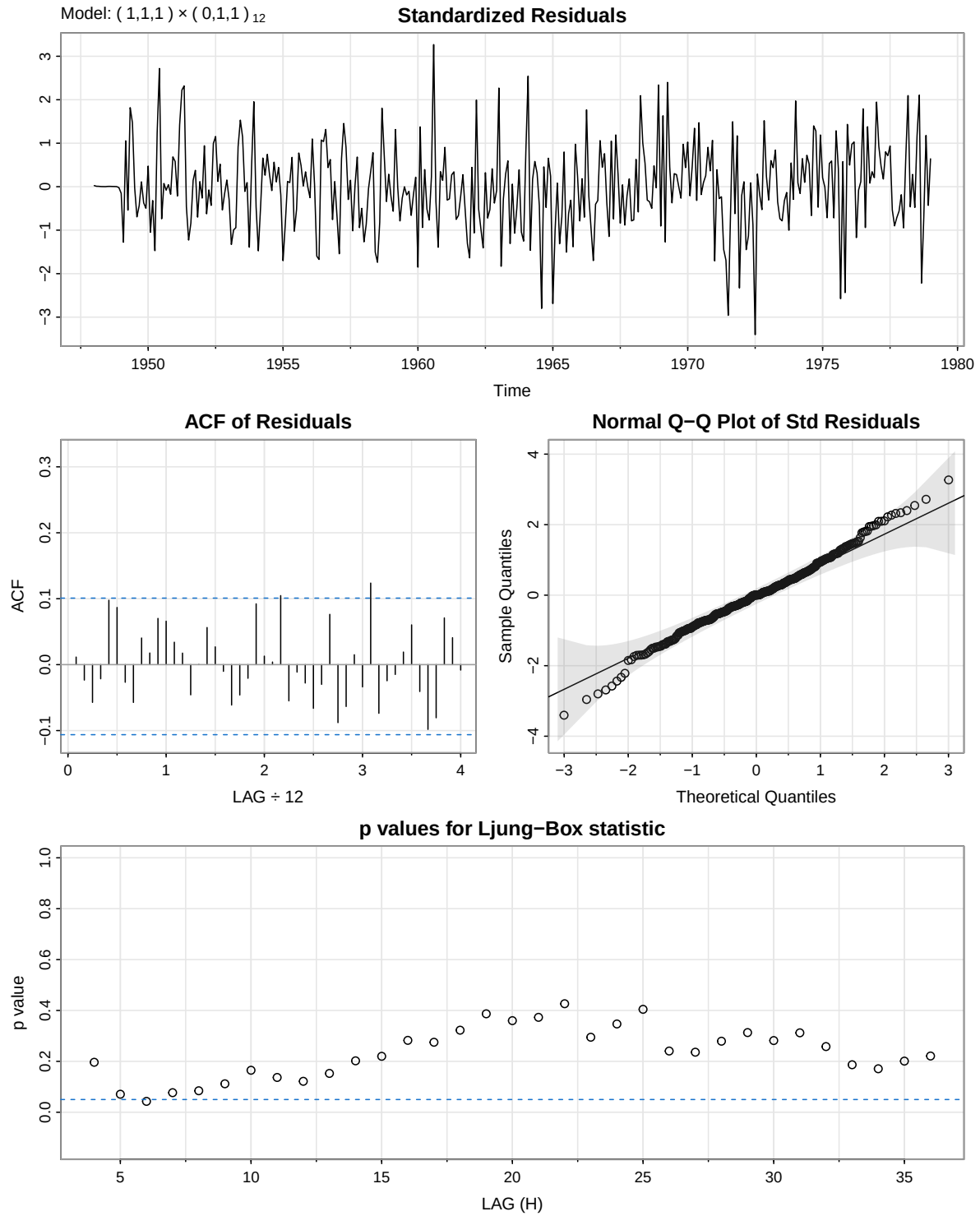
```
initial value 2.218186
iter 2 value 2.032584
iter 3 value 1.982464
iter 4 value 1.975643
iter 5 value 1.971721
iter 6 value 1.967284
iter 7 value 1.963840
iter 8 value 1.961106
iter 9 value 1.960849
iter 10 value 1.960692
iter 11 value 1.960683
iter 12 value 1.960675
iter 13 value 1.960672
iter 13 value 1.960672
iter 13 value 1.960672
final value 1.960672
```

```
initial value 1.940459
iter 2 value 1.934425
iter 3 value 1.932752
iter 4 value 1.931750
iter 5 value 1.931074
iter 6 value 1.930882
iter 7 value 1.930860
iter 8 value 1.930859
iter 8 value 1.930859
final value 1.930859
```

[illegible]

	Estimate	SE	t.value	p.value
ar1	0.3038	0.0865	3.5104	5e-04
ma1	-0.7006	0.0604	-11.5984	0e+00
sma1	-0.8000	0.0441	-18.1302	0e+00

AIC = 6.721818 AIC_c = 6.722006 BIC = 6.764997



The models are comparable except for the p-value for Ljung-Box statistic plot, this model

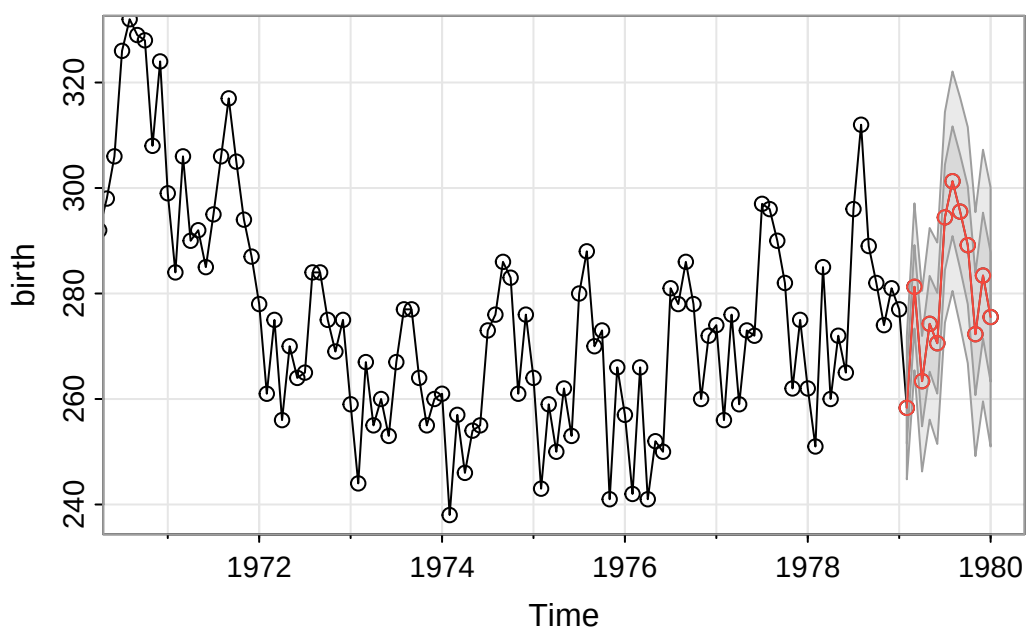
doesn't reject the null hypothesis around lag 7. But generally, the residuals are acceptable.

To make a final decision I compare the AIC/BIC of the two models:

Model	AIC	BIC
$ARIMA(0, 1, 4) \times (0, 1, 1)_{12}$	6.715	6.778
$ARIMA(1, 1, 1) \times (0, 1, 1)_{12}$	6.722	6.765

Using AIC we would select the more complex model, using BIC we would select the simpler model. I choose $ARIMA(1, 1, 1) \times (0, 1, 1)_{12}$ due to it being a simpler model.

The 12 month forecast is below:



\$pred

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
1979	258.3256	281.2730	263.3864	274.2558	270.5740	294.4331	301.2872	
1980	275.5399							
	Sep	Oct	Nov	Dec				
1979	295.5201	289.1691	272.2940	283.4030				
1980								

\$se

	Jan	Feb	Mar	Apr	May	Jun	Jul
--	-----	-----	-----	-----	-----	-----	-----

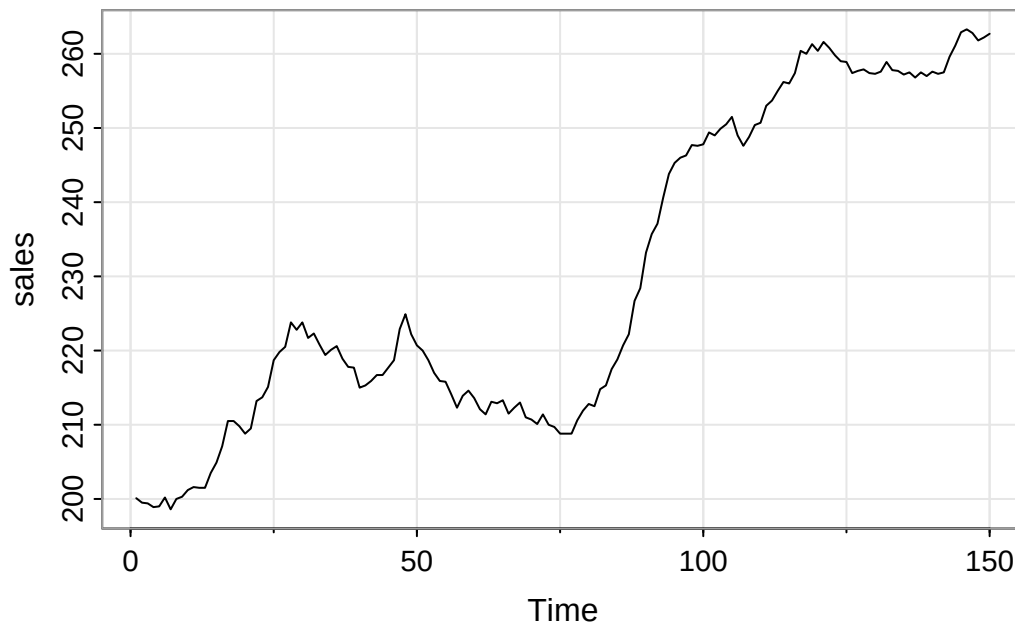
1979		6.776034	7.913388	8.562644	9.080446	9.546734	9.984553
1980	12.274923						
	Aug	Sep	Oct	Nov	Dec		
1979	10.402012	10.802780	11.189036	11.562347	11.923961		
1980							

5.13)

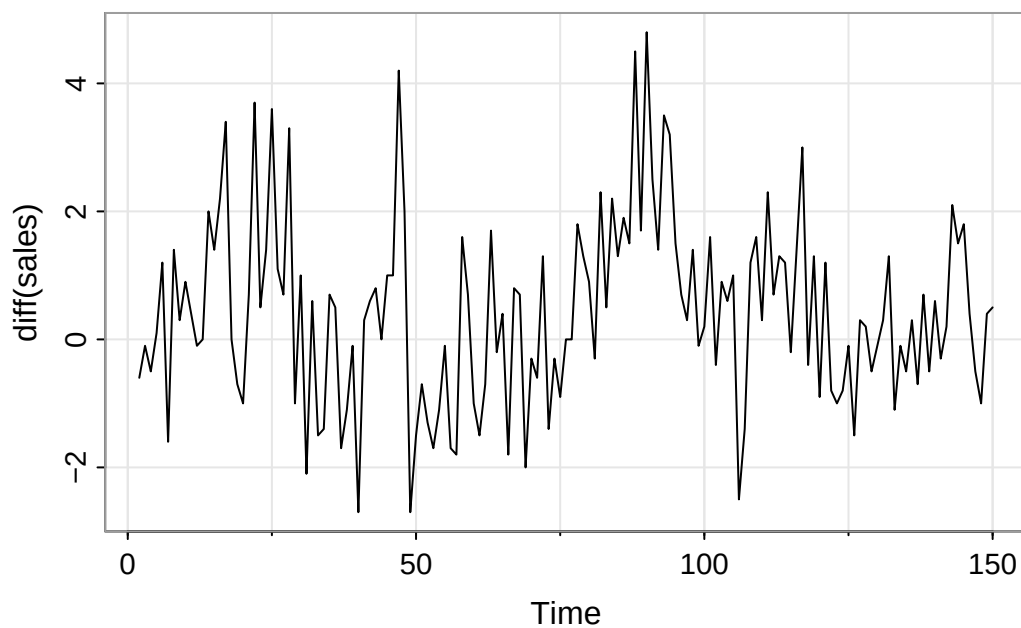
a)

We start by fitting an ARIMA model to the monthly sales data.

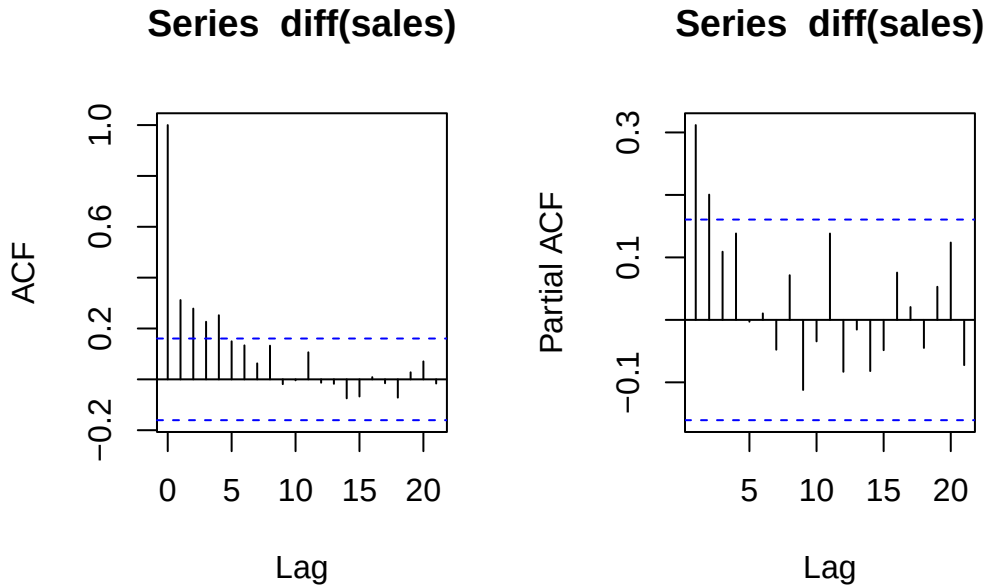
To begin, we look at the time series plot:



This looks like a combination of random walk and overall upward trend. To remedy this we difference:

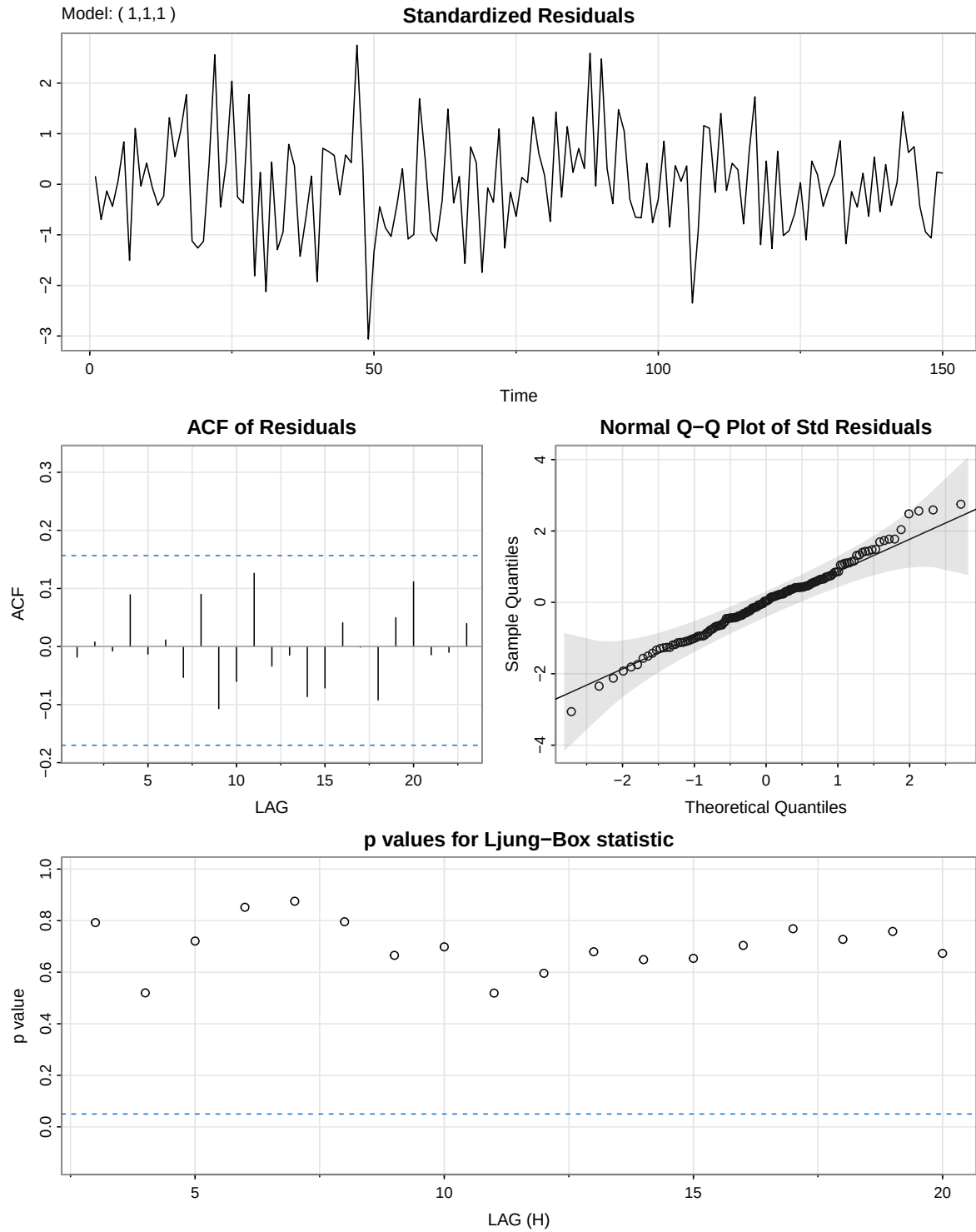


Variation appears constant, and this is a relatively smooth plot as nearby values look positively correlated. I can still see some evidence of upward trend from times 50 to 80 but for now I will look at the ACF and PACF of the data:



The ACF and PACF both seem to tail off, so my first model consideration is $ARIMA(1, 1, 1)$:

AIC = 3.454924 AICc = 3.456035 BIC = 3.535567

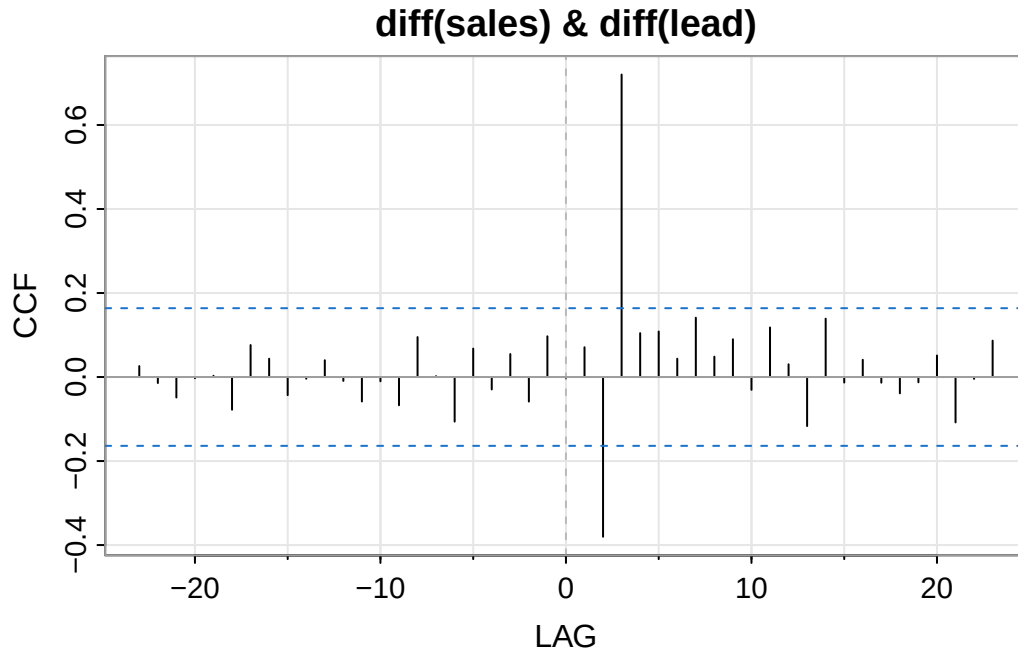


Variance looks constant over time, the ACF and Ljung-Box statistic plot support the assump-

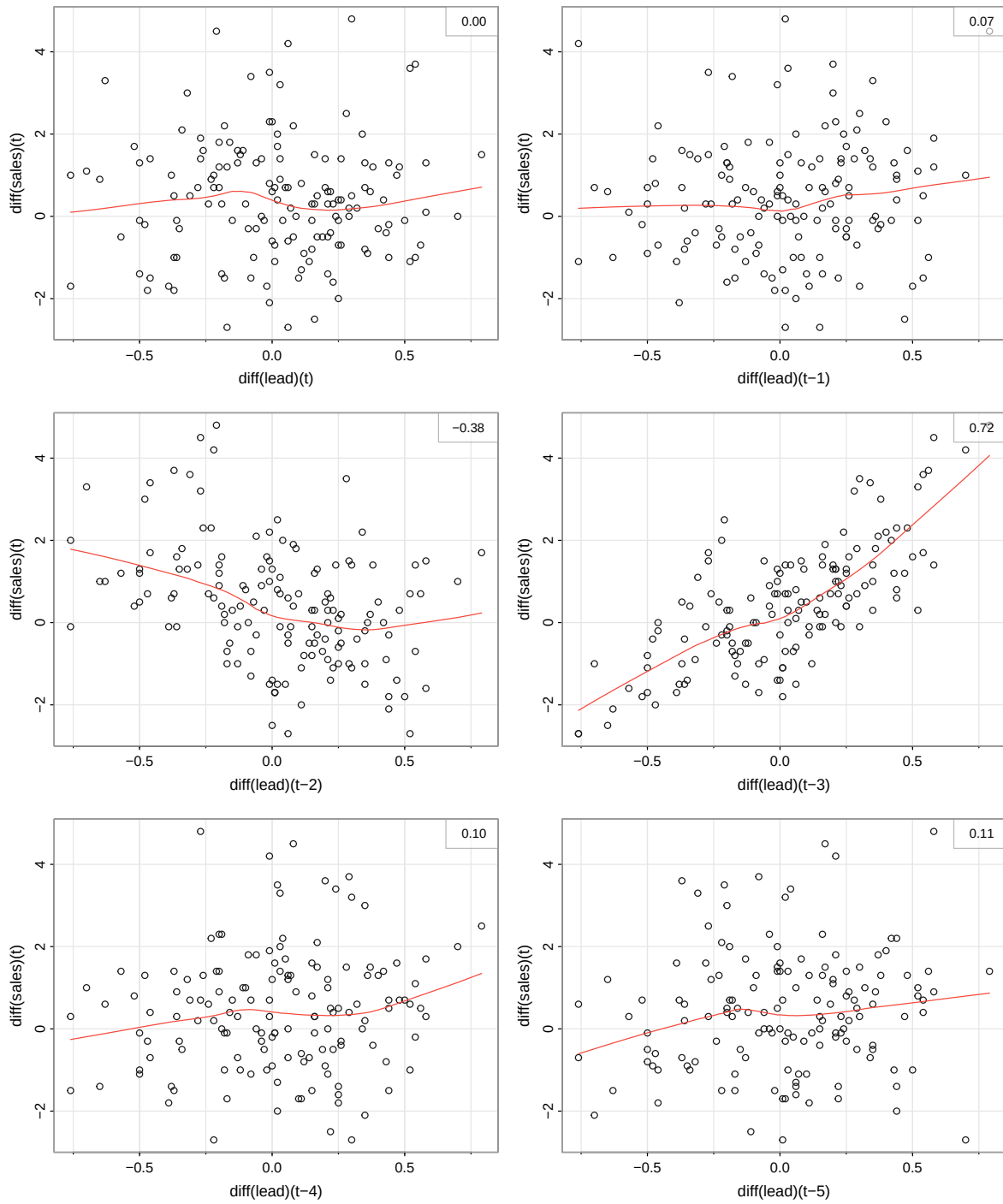
tion of white noise residuals and the residuals look normally distributed. I am satisfied with the $ARIMA(1,1,1)$ model.

b)

We start to investigate the relationship between $\nabla S_t, \nabla L_t$ by using the CCF plot:



The CCF plot shows significant correlation at lags 2 and 3, lets take at the lag plot for the earlier lags.



In this plot we can see the nature of the relationship between the two series. At lag 2, despite having a correlation of -0.38 , the relationship does not look linear. At lag 3 however the

relationship is highly linear and has a strong positive correlation of 0.72. For this reason we should consider adding this to our model.

c)

We are asked to fit $\nabla S_t = \beta_0 + \beta_1 \nabla L_{t-3} + x_t$. To do this we first fit the linear model, $\nabla S_t = \beta_0 + \beta_1 \nabla L_{t-3} + \epsilon$.

```
Attaching package: 'zoo'
```

```
The following objects are masked from 'package:base':
```

```
as.Date, as.Date.numeric
```

```
Call:
```

```
lm(formula = pp$diff_sales ~ pp$diff_leadL3)
```

```
Residuals:
```

Min	1Q	Median	3Q	Max
-2.18876	-0.65502	-0.07291	0.60347	2.84546

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.35538	0.08301	4.281	3.38e-05 ***
pp\$diff_leadL3	3.33733	0.26190	12.743	< 2e-16 ***

```
---
```

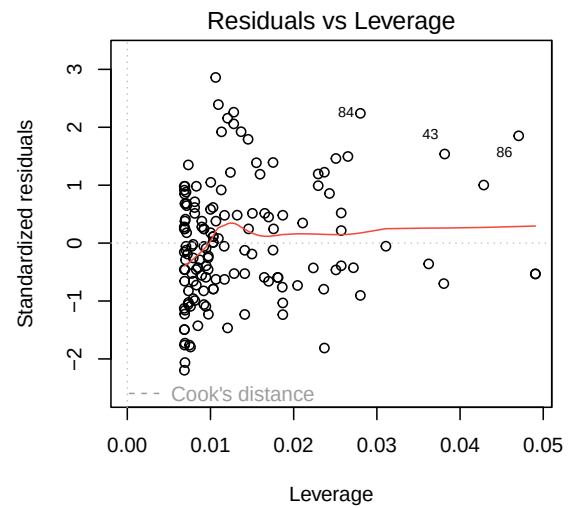
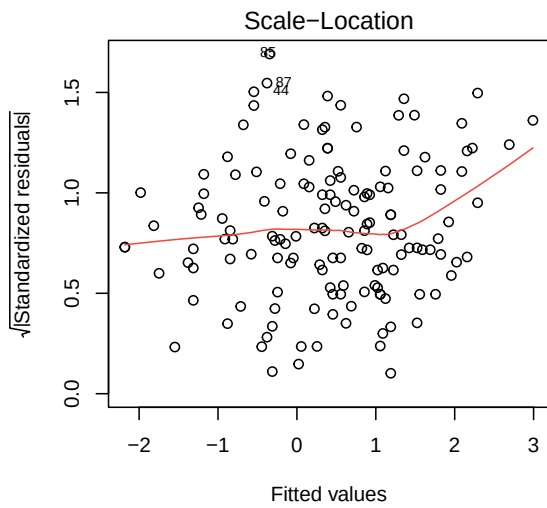
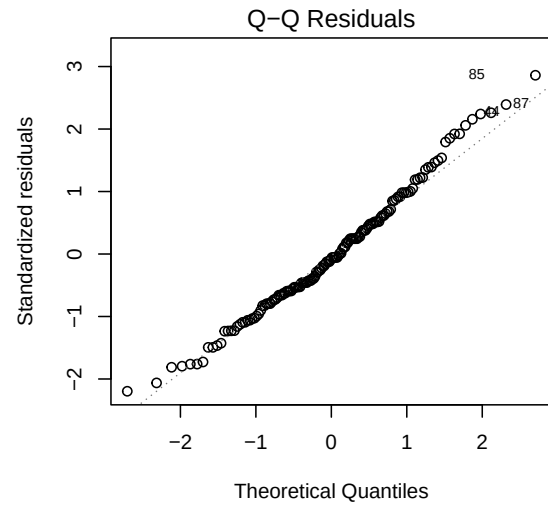
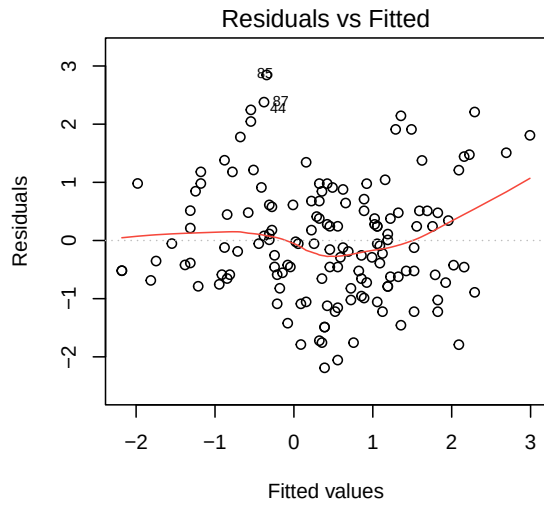
```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1 on 144 degrees of freedom
```

```
Multiple R-squared:  0.53, Adjusted R-squared:  0.5267
```

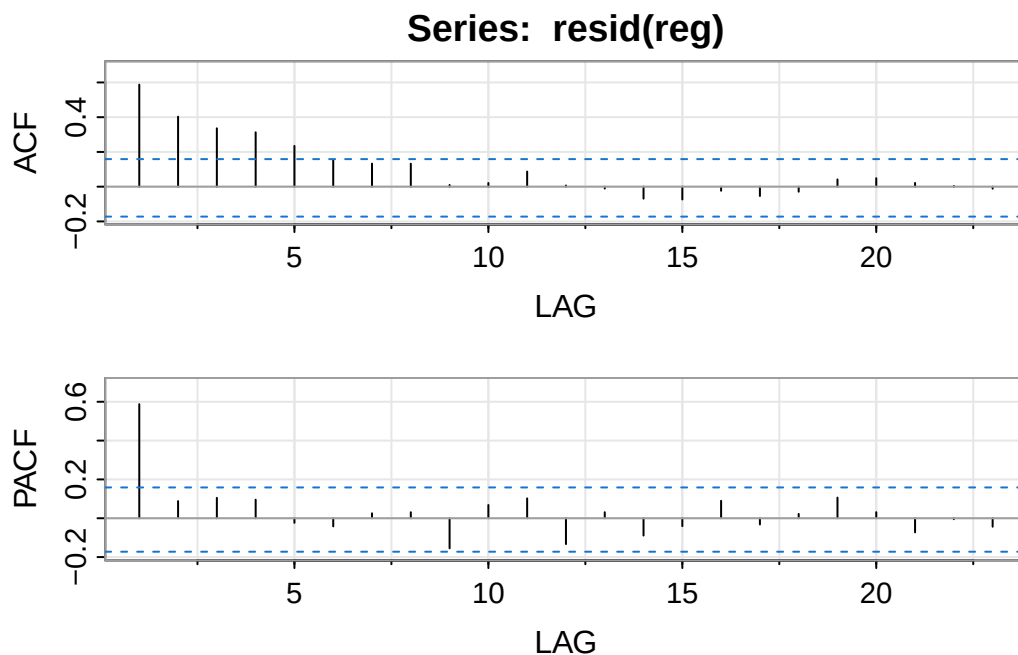
```
F-statistic: 162.4 on 1 and 144 DF,  p-value: < 2.2e-16
```

We can take a quick look at the diagnostic for this simple linear regression:



The only concern would be some evidence of non-constant variance at the right end of the data. Other than that the model is well calibrated.

We take the residuals of that model and use it to fit the ARMA x_t model.



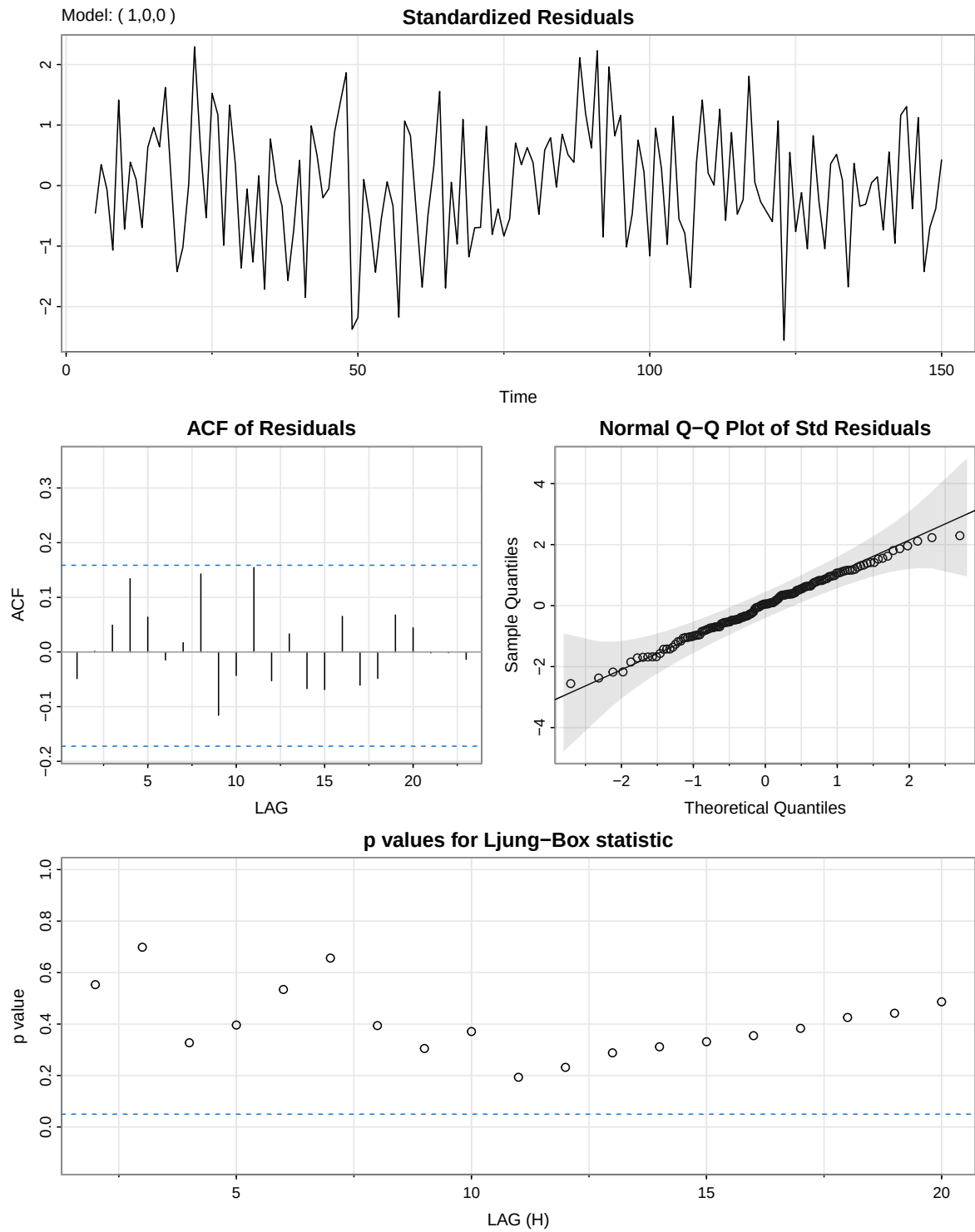
	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]	[,11]	[,12]	[,13]
ACF	0.59	0.40	0.34	0.31	0.23	0.15	0.13	0.13	0.01	0.02	0.09	0.01	-0.01
PACF	0.59	0.09	0.11	0.10	-0.02	-0.04	0.03	0.03	-0.15	0.07	0.10	-0.13	0.03
	[,14]	[,15]	[,16]	[,17]	[,18]	[,19]	[,20]	[,21]	[,22]	[,23]			
ACF	-0.07	-0.07	-0.02	-0.05	-0.03	0.04	0.05	0.02	0.00	-0.01			
PACF	-0.09	-0.04	0.09	-0.03	0.02	0.11	0.03	-0.07	-0.01	-0.04			

The ACF tails off while the PACF cuts to zero after lag one. Using table 4.1 this indicates that we should use $ARMA(1,0)$.

```

initial value -0.004149
iter 2 value -0.216292
iter 3 value -0.216293
iter 4 value -0.216294
iter 5 value -0.216295
iter 5 value -0.216295
iter 5 value -0.216295
final value -0.216295
converged
initial value -0.217539
iter 2 value -0.217552
iter 3 value -0.217561

```

There is no pattern in the standardized residuals, variance looks steady, residuals look normally

distributed and the ACF/Ljung-Box plots both indicate the residuals are white noise. This model fits very well.

We can compare the two models using AIC/BIC:

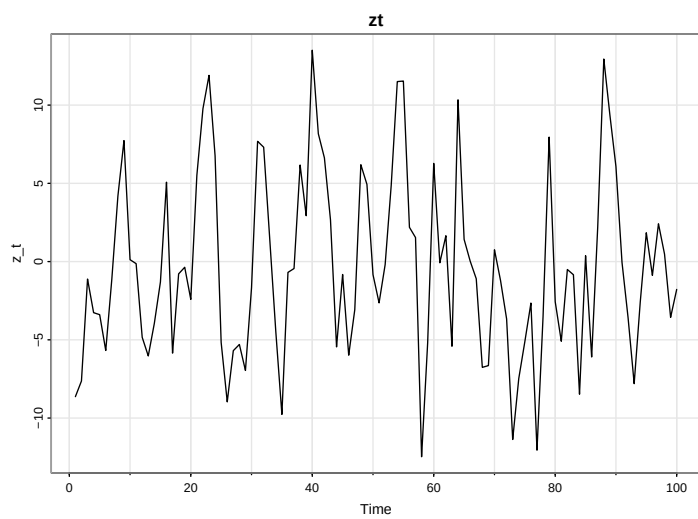
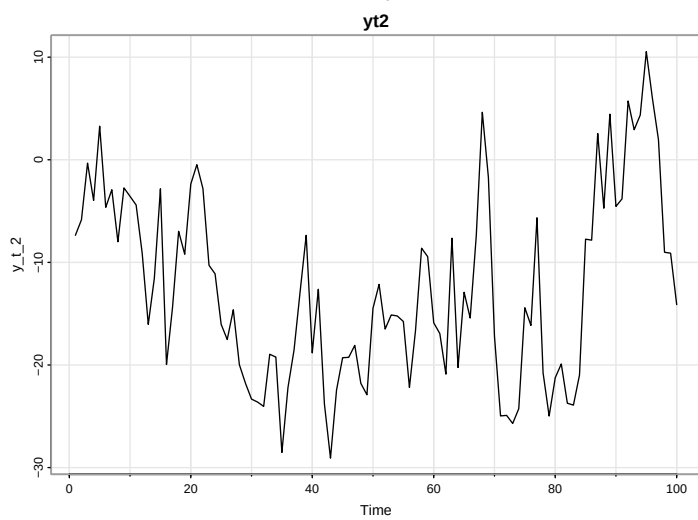
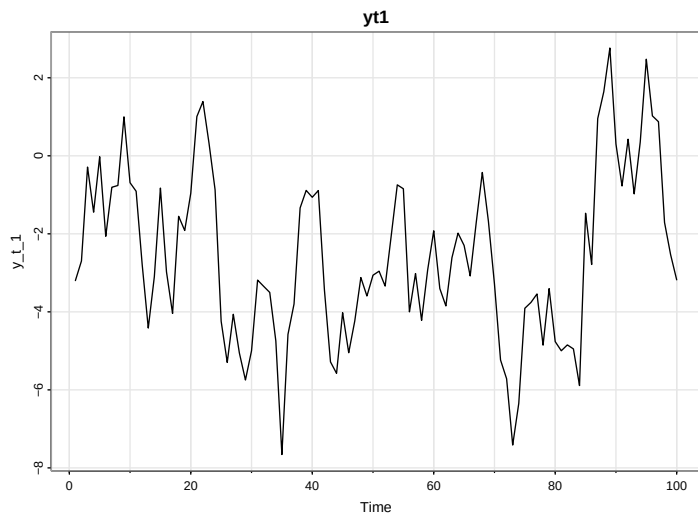
Model	AIC	BIC
ARMA(1,1)	3.455	3.536
AR(1) with Lead Lag 3	2.443	2.505

It is clear that the AR(1) model with the leading indicator is the superior model.

II.

a)

Plotting:



The plots are very smooth which makes me think of random walk and cyclic time series plots, causing me to doubt they are stationary.

b)

For a series to be stationary expected value must be constant and not depend on t and the auto-covariance function must depend only on lag h .

$$E[y_{t1}] = E[x_{t1} + \sum_{j=1}^t w_j] = E[x_{t1}] = E[0.5x_{t-1,1} + w_{t1}] = 0$$

We start with lag = 0 for the auto-covariance function:

$$Cov(y_{t1}, y_{t1}) = Var(y_{t1}) = Var(x_{t1} + \sum_{j=1}^t w_j) = Var(x_{t1}) + t\sigma_w^2 = \frac{1}{1-.5^2} + t$$

Then lag = 1:

$$Cov(y_{t+1,1}, y_{t,1}) = Cov(x_{t+1,1} + \sum_{j=1}^{t+1} w_j, x_t + \sum_{j=1}^t w_j) = Cov(x_{t+1}, x_t) + Cov(\sum_{j=1}^{t+1} w_j, \sum_{j=1}^t w_j) = \gamma(1)_{x1} + t = .5 \frac{1}{1-.5^2}$$

And generally:

$$Cov(y_{t+h,1}, y_{t,1}) = \phi^h \gamma_{x1}(0) + t$$

Where $\gamma_{x1}(0) = \frac{1}{1-\phi^2}$ and $\phi = .5$.

So y_{t1} is not stationary because its autocovariance function depends on t .

c)

We have $z_t = 5y_{t1} - y_{t2}$. Starting with lag = 0:

$$\begin{aligned} Cov(z_t, z_t) &= Var(z_t) = Var(5y_{t1} - y_{t2}) = Var(5(x_{t1} + \sum_{j=1}^t w_j) - (x_{t2} + 5 \sum_{j=1}^t w_j)) = 25Var(x_{t1}) + Var(x_{t2}) \\ &= 25 \frac{1}{1-\phi_{x1}^2} + \frac{1}{1-\phi_{x2}^2} \end{aligned}$$

Where, for ease of notation $\phi_{x1} = .5$ and $\phi_{x2} = .9$.

For lag=1:

$$\begin{aligned} Cov(z_{t+1}, z_t) &= Cov(5y_{t+1,1} - y_{t+1,2}, 5y_{t,1} - y_{t,2}) = Cov(5x_{t+1,t} - x_{t+1,2}, 5x_{t,1} - x_{t,2}) = 25\gamma(1)_{x_1} + \gamma(1)_{x_2} \\ &= 25\phi_{x_1} \frac{1}{1 - \phi_{x_1}^2} \end{aligned}$$

So the autocovariance function is:

$$\gamma(h) = 25\gamma(h)_{x_1} + \gamma(h)_{x_2}$$

Where:

$$\gamma(h)_{x_1} = \begin{cases} \frac{1}{1-.5^2} & h = 0 \\ .5^h \gamma(0)_{x_1} & h \neq 0 \end{cases}$$

and:

$$\gamma(h)_{x_2} = \begin{cases} \frac{1}{1-.9^2} & h = 0 \\ .9^k \gamma(0)_{x_2} & h = 12k, k = 1, 2, .. \\ 0 & \text{otherwise} \end{cases}$$

Since $E[z_t] = E[5y_{t1} - y_{t2}] = 0$, so constant and not depending on t and the auto-covariance function does not depend on t and only on the lag h we can say z_t is a stationary process.

d)

In part b we found that the reason why y_{t1} was non-stationary was due to the sum of the white noise. in part c we found that z_t was able to become stationary because it got rid of the sum of white noise leaving us with the stationary aspects of y_{t1} and y_{t2} .

e)

We start with lag 0 for the cross-covariance function:

$$Cov(y_{t1}, y_{t2}) = Cov(x_{t1} + \sum_{j=1}^t w_j, x_{t2} + 5 \sum_{j=1}^t w_j) = 5t$$

Then with lag 1:

$$Cov(y_{t+1,1}, y_{t,2}) = Cov(x_{t+1,1} + \sum_{j=1}^{t+1} w_j, x_{t,2} + 5 \sum_{j=1}^t w_j) = 5t$$

So when lag is greater than or equal to 0, the covariance is $5t$:

But when lag is less than zero it is a bit more interesting:

$$Cov(y_{t-1,1}, y_{t,2}) = Cov(\sum_{j=1}^{t-1} w_j, 5 \sum_{j=1}^t w_j) = 5(t-1)$$

So the cross-covariance function is:

$$Cov(y_{t+h,1}, y_{t,2}) = \begin{cases} 5t & h \geq 0 \\ 5(t-h) & h < 0 \end{cases}$$

The cross-correlation function is:

$$\rho_{xy}(y_{t+h,1}, y_{t,2}) = \frac{Cov(y_{t+h,1}, y_{t,2})}{\sqrt{\gamma_{y_1}(0)\gamma_{y_2}(0)}}$$

The denominator would be:

$$\sqrt{(\frac{1}{1-.5^2} + t)(\frac{1}{1-.9^2} + t)}$$

The two time series are not jointly stationary because their cross-covariance function depends on t .